

Interdisciplinary Guide — T1/T4 & InterIA

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Reading routes (by specialty)

NT. $T1' \rightarrow T2 \rightarrow T3$; $T4$ minimal.

AO. $T2 \rightarrow T1' \rightarrow T3$; $T4$ as spectral reading.

Signal. Tapers, bandlimit; $T2$ as frame conditioning; $T3$ as matched-filter probe.

Phys. Rigidity ($T2$) + localized measurement ($T3$); HP compatible.

Repro. See InterIA (separate).

Formal. Lemmas: Schur/Gershgorin, PW density, Archimedean, absorption.

Notations and preliminaries

NT	Cross reading
Band $[-A, A]$	Bandwidth (signal)
Cos^2 taper	Apodization (plateau/transition)
Ξ_A	Non-uniform sampling
$\log p/p^{k/2}$	p -adic energy weighting
$C_{\text{fr}}^{(w)} \geq 1 - \Theta_w$	Frame conditioning (AO)
Wave-packet $\hat{\phi}_T$	Matched filter (signal/phys)
Explicit formula	Source-spectrum coupling

Storyboard T1/-T4

T1/: set the arena (PW, explicit, Archimedean).

T2: gap via locality + weights.

T3: packet $\rightarrow$ contradiction if off-line zero.

T4: HP-compatible reading (no postulate).

Bridges boxes

(AO) $G^{(w)}$ Gram; $I + R$ with $\|R\| < 1$ gives lower boundedness.

(Signal) Band $[-A, A]$, apodized taper, frame $\approx$ robust reconstruction.

(Phys) T2 = spectral rigidity; T3 = localized measurement.

(Repro) Interval certificates; proof does not depend on numerics.

Micro-example (T2)

$A = 4; p \leq e^4.$

Few neighbors (locality), decay $p^{-(k-k_\alpha)/4}.$

Indicative: $\Theta_w(A) \approx 0.63 \Rightarrow C_{\text{fr}}^{(w)} \approx 0.37 > 0.$

Dependency map

